

Software Requirements Specification

Bus and Train Reservation System

Topic 8

DBMS Project Document

1. Introduction

1.1 Purpose

The purpose of this SRS is to define the requirements for a Bus and Train Reservation System. The document is written for a DBMS project, so the main attention is on stored data, relationships, constraints, transactions, and reports.

1.2 Scope

The system is used by a regional transport operator that runs scheduled bus and train services on fixed routes. Passengers can search services between two stops, book seats for a future service run, pay online, receive an e-ticket, and cancel tickets according to the refund policy.

The system shall maintain route details, stops, service runs, passengers, identification documents, bookings, tickets, seats, payments, cancellations, refunds, and reports. Detailed screen design and user interface layout are outside the scope of this document.

2. Overall Description

2.1 Product Perspective

The system works as a central reservation database for bus and train services. It stores master data such as routes, stops, services, transports, and seats. It also stores transaction data such as bookings, tickets, payments, cancellations, and refunds.

The database must support leg-wise seat availability. A seat already booked from City A to City B cannot be booked again for any overlapping segment. The same seat may be booked from City B to City C if the earlier passenger gets off at City B.

2.2 User Roles

User Role	Main Responsibility
Passenger	Searches services, books tickets, pays online, downloads e-ticket, and cancels tickets.
Booking Clerk	Books tickets for passengers and checks booking or cancellation details.
Operations Staff	Manages route, stop, service, transport, and service run data.
Administrator	Manages users, roles, system data, and reports.

2.3 Assumptions

- All routes and stops are entered before generating bookable service runs.
- Payment status is updated before e-ticket generation.
- Each passenger must have a user account for online booking.
- Identification document details are stored when required by booking rules or concession rules.
- Date and time values are stored in a consistent format in the database.

3. Functional Requirements

ID	Requirement	Description
FR-01	Route and stop management	The system shall store routes with route name and total distance. It shall store ordered route stops with stop sequence, distance from origin, arrival time, and departure time.
FR-02	Service definition	The system shall store bus or train service details, service type, status, assigned route, transport, and seat capacity.
FR-03	Service run generation	The system shall create date-wise service runs from the defined service and transport data. Each service run shall have departure time, arrival time, and status.
FR-04	Service search	The system shall search service runs based on source stop, destination stop, and travel date. Only valid runs where the source stop comes before the destination stop shall be shown.
FR-05	Seat selection	The system shall check leg-wise seat availability before booking. A seat cannot be selected if an existing ticket has an overlapping segment for the same service run and same seat.
FR-06	Booking creation	The system shall create a booking record with passenger, booking date and time, total amount, booking status, payment reference, identification document reference, and boarding/disembarking stop references.

FR-07	Payment recording	The system shall store payment date and time, amount, mode, status, and transaction ID. A booking shall be confirmed only after successful payment.
FR-08	Ticket generation	The system shall generate ticket records with fare, boarding information, disembarking information, booking reference, and service run reference.
FR-09	E-ticket issue	The system shall issue an e-ticket after successful booking. The e-ticket shall store issue date and time, QR code, PDF URL, and related ticket ID.
FR-10	Cancellation	The system shall store cancellation details with reason, cancellation date and time, and ticket reference.
FR-11	Refund	The system shall store refund amount, refund date and time, status, and cancellation reference. Refund amount shall be calculated based on the cancellation policy.
FR-12	Reports	The system shall generate route-wise occupancy and revenue reports from bookings, tickets, service runs, routes, and payment records.

4. Business Rules and Constraints

Rule ID	Rule
BR-01	A seat sold for an overlapping segment of the same service run cannot be sold again.
BR-02	A seat may be sold for a later non-overlapping segment of the same service run.
BR-03	Cancellation charges depend on how far before departure the ticket is cancelled.
BR-04	A passenger can hold at most six tickets per identification document per service.
BR-05	Concession fares require a valid supporting document to be recorded.
BR-06	A ticket cannot be issued unless the related booking and payment are valid.
BR-07	A refund record can be created only for a valid cancellation record.

4.1 Seat Overlap Rule

For a selected seat, source stop, destination stop, and service run, the system shall compare the selected route segment with already confirmed tickets. The new booking is rejected when the selected segment overlaps with an existing booked segment for the same seat and service run.

The overlap condition can be checked using stop sequence numbers:

- $\text{new_boarding_sequence} < \text{existing_destination_sequence}$, and
- $\text{new_destination_sequence} > \text{existing_boarding_sequence}$.

If both conditions are true, the two journeys overlap and the seat is not available.

5. Database Design

5.1 ER Diagram

The ER diagram shows the main entities and relationships used in the reservation database.

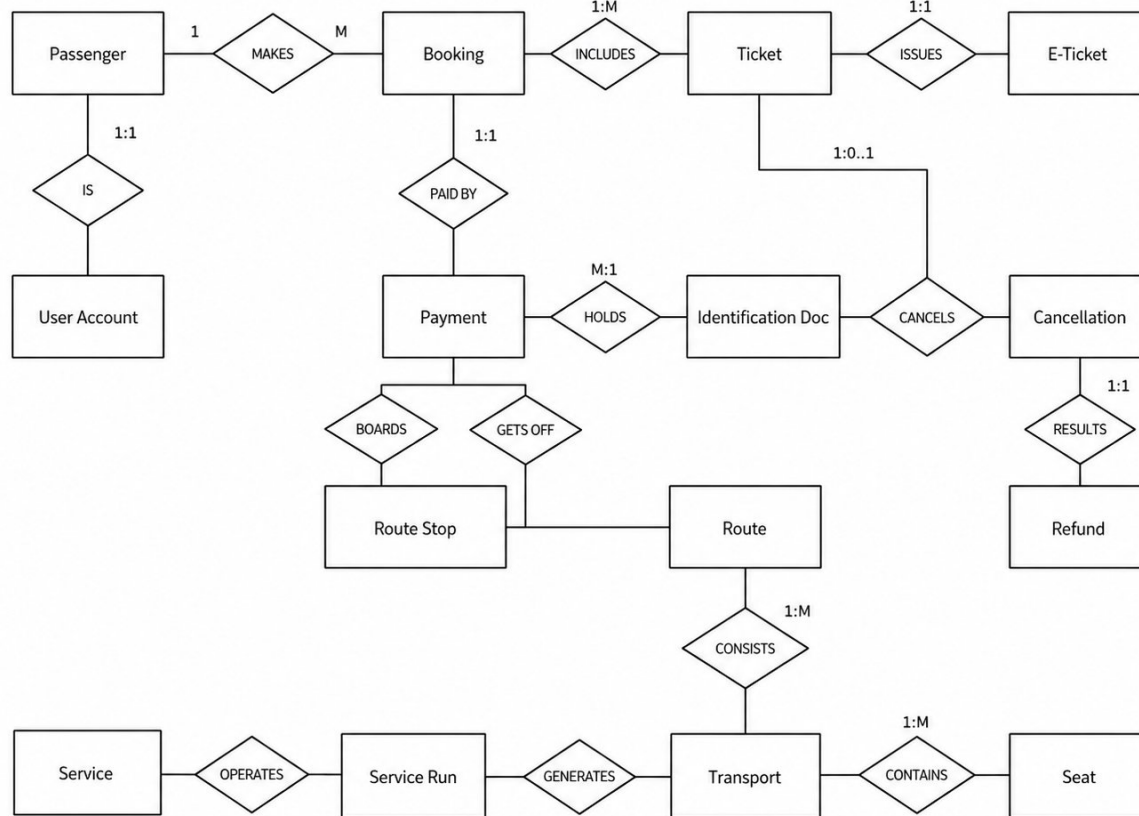


Figure 1: ER Diagram of Bus and Train Reservation System

5.2 Relational Schema

The following relational schema is used for the database design.

PASSENGER(p_id PK, name, phone, email, gender, age, user_id FK)

USER_ACCOUNT(user_id PK, name, password_hash, full_name, phone, role_id FK)

ROLE(role_id PK, role_name)

BOOKING(booking_id PK, date_time, total_amount, status, p_id FK, payment_id FK, document_id FK, boarding_stop_id FK, disembarking_stop_id FK)

IDENTIFICATION_DOCUMENT(document_id PK, doc_type, doc_url)

PAYMENT(payment_id PK, date_time, amount, mode, status, transaction_id)

TICKET(ticket_id PK, fare_amount, boarding_info, disembarking_info, booking_id FK, service_run_id FK)

E_TICKET(e_ticket_id PK, issue_datetime, qr_code, pdf_url, ticket_id FK)

CANCELLATION(cancellation_id PK, reason, date_time, ticket_id FK)

REFUND(refund_id PK, amount, date_time, status, cancellation_id FK)

SERVICE_RUN(service_run_id PK, run_id, departure_time, arrival_time, status, transport_id FK, service_id FK)

SERVICE(service_id PK, service_name, service_type, status, route_id FK)

TRANSPORT(transport_id PK, transport_type, transport_number, capacity)

SEAT(seat_id PK, seat_no, transport_id FK)

ROUTE(route_id PK, route_name, total_distance)

ROUTE_STOP(route_stop_id PK, stop_sequence, distance_from_origin, arrival_time, departure_time, route_id FK)

5.3 Data Dictionary

Table	Purpose	Primary Key	Foreign Key(s)
PASSENGER	Stores passenger personal details.	p_id	user_id
USER_ACCOUNT	Stores login account details and role reference.	user_id	role_id

ROLE	Stores role names such as passenger, clerk, operations staff, and administrator.	role_id	-
ROUTE	Stores fixed route information.	route_id	-
ROUTE_STOP	Stores stops under a route in travel order.	route_stop_id	route_id
SERVICE	Stores service information for a route.	service_id	route_id
SERVICE_RUN	Stores actual bookable run of a service.	service_run_id	service_id, transport_id
TRANSPORT	Stores bus or train vehicle details.	transport_id	-
SEAT	Stores seat numbers for a transport.	seat_id	transport_id
BOOKING	Stores booking transaction header.	booking_id	p_id, payment_id, document_id, boarding_stop_id, disembarking_stop_id
PAYMENT	Stores online payment details.	payment_id	-
TICKET	Stores ticket details for a booking and service run.	ticket_id	booking_id, service_run_id
E_TICKET	Stores digital e-ticket details.	e_ticket_id	ticket_id
IDENTIFICATION_DOCUMENT	Stores supporting document details.	document_id	-
CANCELLATION	Stores ticket cancellation details.	cancellation_id	ticket_id
REFUND	Stores refund information for a cancellation.	refund_id	cancellation_id

5.4 Main Relationships

Relationship	Meaning
Passenger - Booking	One passenger can make many bookings.
Booking - Payment	Each booking is paid by one payment record.
Booking - Ticket	A booking includes ticket records.
Ticket - E Ticket	A ticket has an issued e-ticket.
Ticket - Cancellation	A ticket may have a cancellation record.
Cancellation - Refund	A cancellation results in one refund record when applicable.
Route - Route_Stop	A route consists of ordered route stops.
Route - Service	A service belongs to a route.
Service - Service_Run	A service generates bookable service runs.
Transport - Seat	A transport contains seats.

6. Database Constraints and Transaction Rules

6.1 Entity Integrity

- Every table shall have a primary key.
- Primary key values shall be unique and not null.
- Important fields such as passenger name, route name, service status, and payment status shall not be left empty.

6.2 Referential Integrity

- A booking shall not exist without a valid passenger.
- A ticket shall not exist without a valid booking and service run.
- A service run shall not exist without a valid service and transport.
- A seat shall not exist without a valid transport.
- A refund shall not exist without a valid cancellation.

6.3 Domain Constraints

Data Item	Constraint
Payment status	Pending, Successful, Failed, Refunded
Booking status	Pending, Confirmed, Cancelled
Service type	Bus or Train
Service run status	Open, Closed, Cancelled
Amount fields	Must be zero or positive.
Stop sequence	Must be positive and unique within the same route.

6.4 Transaction Handling

Booking confirmation shall be handled as a transaction. Seat availability check, booking insert, ticket insert, payment status update, and e-ticket issue must be completed together. If any step fails, the transaction shall be rolled back to avoid partial booking data.

Cancellation and refund creation shall also be handled as one transaction so that cancelled tickets and refund status remain consistent.

7. Reports

Report	Data Used
Route-wise occupancy report	Shows booked seats or tickets for each route and service run.
Revenue report	Shows received amount based on successful payment records.
Cancellation report	Shows cancelled tickets and cancellation reasons.
Refund report	Shows refund amount and refund status.

8. Non-Functional Requirements

Category	Requirement
Performance	Seat availability and service search should return results without long delay for normal usage.
Security	Passwords shall be stored as hashes. Role-based access shall be used for administration and operations tasks.
Reliability	Confirmed bookings shall not be lost after payment success. Important operations shall use database transactions.
Backup	The database shall be backed up regularly to protect booking and payment data.
Data Accuracy	Fare amount, payment amount, cancellation charge, and refund amount shall be stored accurately.
Auditability	Booking time, payment time, cancellation time, and refund time shall be stored for checking records later.

9. Use Case Summary

Use Case ID	Use Case	Actor	Result
UC-01	Manage Route and Stops	Operations Staff	Route and route stop records are created or updated.
UC-02	Generate Service Run	Operations Staff	A service run becomes available for booking.
UC-03	Search Service	Passenger / Clerk	Available service runs are displayed for the given date and stops.
UC-04	Book Ticket	Passenger / Clerk	Booking, ticket, payment, and e-ticket records are created.
UC-05	Cancel Ticket	Passenger / Clerk	Cancellation and refund records are created according to policy.
UC-06	View Reports	Administrator	Occupancy and revenue reports are generated from database records.

10. Acceptance Criteria

1. The database stores all master data: route, route stop, service, service run, transport, and seat.
2. The system can search services between two valid stops on a route for a selected date.
3. The system prevents double booking of a seat for overlapping route segments.
4. Booking, payment, ticket, and e-ticket records are correctly linked by foreign keys.
5. Cancellation and refund records are correctly linked to tickets.
6. The database can produce route-wise occupancy and revenue reports.
7. User roles control access to passenger, booking clerk, operations, and administrator functions.